

§11.10 Taylor and Maclaurin Series

Homework : 3,17,23,31,39,45,47,57,62

Questions :

- (i) How to find a power series representation at a of a function f in general?
- (ii) Will such power series representation equal to the function f itself?

General Method: If $f(x)$ has a power series representation at a , say,

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n, \quad |x-a| < R, \quad \text{then} \quad \boxed{c_n = \frac{f^{(n)}(a)}{n!} .}$$

Definition :

- (i) Taylor series of f at $x=a$ is defined to be $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$.
- (ii) If $a=0$, the corresponding Taylor series is called Maclaurin series.

Remark : $f(x)$ is not necessarily equal to its Taylor series.

Example 1 : $f(x) = \begin{cases} e^{-\frac{1}{x^2}} & \text{if } x \neq 0. \\ 0 & \text{if } x = 0. \end{cases}$

Solution :

$$\Rightarrow f'(0) = f''(0) = \dots f^{(n)}(0) = 0$$

$$\Rightarrow f(x) \neq \text{Maclaurin series} = 0 \text{ but } f(x) \neq 0 \text{ whenever } x \neq 0.$$

Example 2 : Let $f(x) = e^x$. Find its Maclaurin series.

Solution :

$$\text{M.S.} = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \frac{x^n}{n!} \cdot (\because f^{(n)}(x) = e^x).$$

Example 3 : Let $f(x) = \sin x$. Find its Maclaurin series.

Solution :

$$\begin{aligned} f'(x) &= \cos x \\ f''(x) &= -\sin x \\ f^{(3)}(x) &= -\cos x \\ f^{(4)}(x) &= \sin x \end{aligned} \Rightarrow \text{M.S.} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}.$$

Example 4 : $f(x) = \cos x$

Solution :

$$\Rightarrow \text{M.S.} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{2n!}.$$

誤差
↑

Taylor Inequality (誤差估計) : Let $f(x) = T_n(x) + R_n(x)$, where

$T_n = \sum_{i=0}^n \frac{f^{(i)}(0)}{i!} (x-a)^i$ is called the n th-degree Taylor polynomial

of f at a . If $|f^{(n+1)}(x)| \leq M$ for $|x-a| \leq d$, then

$$|R_n(x)| \leq \frac{M}{(n+1)!} |x-a|^{n+1} \text{ for } |x-a| \leq d$$

Example 5 : Prove $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ for any x .

Solution : $|R_n(x)| \leq \frac{e^d}{(n+1)!} |x|^{n+1} \rightarrow 0$ as $n \rightarrow \infty$ for any x .

註：對任一 $x \in R$, $\sin x$ and $\cos x$ 的 Maclaurin series 分別等於 $\sin x$ and $\cos x$.

Example 6 : $3 + \frac{9}{2!} + \frac{27}{3!} + \frac{81}{4!} + \dots$

Solution :

$$3 + \frac{9}{2!} + \frac{27}{3!} + \frac{81}{4!} + \dots$$

$$= e^3 - 1 .$$

Example 7 : $\frac{1}{2!} - \frac{1}{4!} + \frac{1}{6!} - \frac{1}{8!} + \dots$

Solution :

$$\frac{1}{2!} - \frac{1}{4!} + \frac{1}{6!} - \frac{1}{8!} + \dots$$

$$= 1 - \cos 1 .$$